Functions and 2D lists Assignment (/18 marks)
(17 marks + 1 comments)

<u>In all 4 questions make sure to test the functions!!!</u>

1. Create a function called **countVowels** that takes a string and returns the number of vowels (a, e, i, o, u) in the string. (it should count both capital and lowercase vowels)

   (/3 marks)

2. In a game of chess, a bishop moves diagonally any number of squares. Create a function in Python called **bishopMoves** that takes in the current location (row, col) of a bishop on the chess board and returns a list of tuples (row, col) of all possible places the bishop can move. (/5 marks)

(both row and col can be any number between 1 and 8 (not 0 to 7))

3.Write a function named **checkerboard** that will accept 2 integers (m and n) and will **return a 2D list** of size (m×n) and populate it with 1 and 0 in a checkerboard pattern. The top left corner should start with 1.

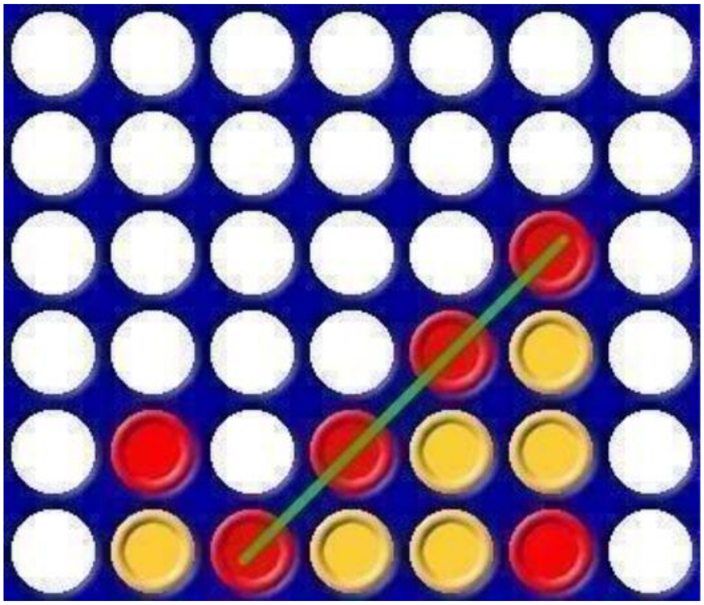
checkerboard(3,4) will create the following 2D list

[[1, 0, 1, 0],
 [0, 1, 0, 1],
 [1, 0, 1, 0]]

e.g. checkerboard(6,8) will create the following 2D list (4 marks)

[[1, 0, 1, 0, 1, 0, 1, 0],
 [0, 1, 0, 1, 0, 1, 0, 1],
 [1, 0, 1, 0, 1, 0, 1, 0],
 [0, 1, 0, 1, 0, 1, 0, 1],
 [1, 0, 1, 0, 1, 0, 1, 0],
 [0, 1, 0, 1, 0, 1, 0, 1]]

4. Create a function called **connect4** that takes a 6x7 list of integers where the values will be 0, 1 or 2, and returns **True** if there are four ones or four twos in a row, horizontally, vertically or diagonally and returns **False** otherwise.



(5 marks)